

ALICIA M. ANDERSON

RESEARCH STATEMENT

Astrophysics PhD candidate at the University of Cambridge specializing in high-resolution spectroscopy and astronomical instrumentation. My current research focuses on mitigating instrumental and pipeline systematics in precision radial velocity measurements with the high-resolution cross-dispersed echelle spectrograph, HARPS3. I have experience in cryogenic instrumentation, optical design and assembly, and the development of custom analysis and simulation tools for spectroscopy. Proficient in Python, C, and C++, with a strong background in scientific software and hardware integration.

EDUCATION

Trinity College, University of Cambridge, United Kingdom *October 2022 - present*
Ph.D. in Physics
Thesis: *Minimising instrumental and pipeline systematics to achieve extreme-precision radial velocity measurements with HARPS3*
Supervisors: Dr. Samantha Thompson, Dr. Vinesh Rajpaul
PI: Prof. Didier Queloz

University of Lethbridge, Canada *June 2022*
Master of Science in Physics
Thesis: *Development of the dispersing component for a cryogenic post-dispersed polarizing Fourier transform spectrometer*
GPA: 4.0/4.0
Bachelor of Science in Physics with Honours and Great Distinction *June 2020*
Graduation GPA: 4.0/4.0

RESEARCH EXPERIENCE

- University of Cambridge** *October 2022 – Present*
PhD Student in Astrophysics *Cambridge, United Kingdom*
- Contributing to the assembly, integration, and verification of HARPS3 — a high-resolution cross-dispersed spectrograph designed for extreme-precision radial velocity measurements.
 - Developing software to simulate the HARPS3 spectral format for performance validation.
 - Assisting in the integration and early validation of the HARPS3 data reduction pipeline, with a focus on optimization and minimising systematics.
- University of Cambridge** *November 2021*
Research Assistantship *Cambridge, United Kingdom*
- Completed a 6-week internship studying high-resolution interferometric techniques to probe the intrinsic absorption modes of an InGaAs detector at infrared wavelengths.
- University of Lethbridge Astronomical Instrumentation Group** *May 2022 - September 2022*
Research Internship *Lethbridge, Canada*
- Designed and operated laboratory experiments for the validation of a cryogenic far-IR spectrometer at 4 K.
 - Collected and analyzed data for several publications.
- Blue Sky Spectroscopy** *September 2019 - April 2020*
Research Internship *Lethbridge, Canada*
- Conducted 4 K validation experiments for cryogenic far-IR spectrometer.
- University of Lethbridge** *May - August 2019*
NSERC Undergraduate Student Research Award *Lethbridge, Canada*

- Developed cryogenic instrumentation for far-infrared astronomical applications.

University of Calgary

NSERC Undergraduate Student Research Award

May - August 2018

Calgary, Canada

- Applied high-resolution spectroscopy to study Van der Waals forces in molecular clusters.

University of Lethbridge

NSERC Undergraduate Student Research Award

May - August 2017

Lethbridge, Canada

- Investigated the application of Fourier transform spectroscopy to astronomical imaging.

TEACHING EXPERIENCE

University of Cambridge

Undergraduate Supervisor

January 2023 - present

Cambridge, United Kingdom

- Supervising students in the Physics Part III minor topic: Exoplanets and Planetary Systems.
- Assisting undergraduate students with laboratory experiments in the Physics Part IA course.

University of Lethbridge

Graduate Teaching Assistant

September 2020 - May 2022

Lethbridge, Canada

- Instructed laboratory sessions for undergraduate introductory physics courses.
- Ran tutorial sessions for introductory physics.

AWARDS AND HONOURS

University of Cambridge Awards

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| 2024 | Lindau Nobel Laureate Meeting (Physics) — Selected delegate |
| | Canadian Centennial Scholarship Foundation Award |
| 2022–2026 | Gates Cambridge Scholarship |
| | Natural Sciences and Engineering Research Council of Canada Doctoral Award |
| | Trinity College Krishnan-Ang Studentship in the Natural Sciences — <i>Honorary</i> |
| | Cambridge International Scholarship — <i>offered, declined</i> |

University of Lethbridge Awards

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| 2022 | Graduate Studies Silver Medal of Merit |
| 2020–2022 | NSERC CGS-M and NTCO Master's Awards |
| | Métis Scholar Graduate Award |
| 2020 | Faculty of Arts & Science Undergraduate Gold Medal — <i>nominated</i> |
| 2017–2019 | NSERC Undergraduate Student Research Awards (3x) |
| 2018–2020 | Indigenous Careers Award |
| 2019/20 | Jason Lang Scholarship |
| 2018/19 | Joe Rood Memorial Award — Top student in Physics |
| 2016/17 | Rogers Sugar Scholarship — Highest GPA in Physics & Chemistry |
| | Louise McKinney Scholarship |
| 2015–2020 | Dean's Honour List |
| | Academic All-Canadian |
| | Athletic Scholarship |

Peer-reviewed Journal Articles

Buchan, M. A., Naylor, D. A., Gom, B. G., Christiansen, A. J., **Anderson, A. M.**, Cournoyer, A., Carbonneau, E., Larouche, M., Bourque, H. A., Ade, P. A. R., Tucker, C., "Design and end-to-end validation of a cryogenic far-infrared post-dispersed polarizing Fourier transform spectrometer," *Journal of Astronomical Telescopes, Instruments, and Systems*, Vol. 11, Issue 3, 031640, 2025, <https://doi.org/10.1117/1.JATIS.11.3.031640>.

Anderson, A. M., Naylor D. A., Gom B. G., Buchan, *et. al.* "Development and validation of a cryogenic far-infrared diffraction grating spectrometer used to post-disperse the output from a Fourier transform spectrometer." *Review of Scientific Instruments*, vol. 95, no. 1, 2024, <https://doi.org/10.1063/5.0177603>.

Conference Proceedings

Buchan, M. A., Naylor, D. A., Gom, B. G., **Anderson, A. M.**, *et. al.* "Development and Integration of a Cryogenic Far-infrared Post-dispersed Polarizing Fourier Transform Spectrometer." *Optica Sensing Congress 2023 (AIS, FTS, HISE, Sensors, ES)*, 2023, <https://doi.org/10.1364/fts.2023.fm4b.4>.

Anderson, A. M., Naylor, D. A., Gom, B. G., Veenendaal, I. T., *et. al.* "Determining the Efficiency of a Cryogenic Far-infrared Diffraction Grating Spectrometer Used as a Post-dispersing Module for a High-resolution Spectrometer." *Proc. SPIE 12190*, 2022, <https://doi.org/10.1117/12.2628259>.

Naylor, D. A., Gom, B. G., **Anderson, A. M.**, Huber, A. I., *et. al.* "Development of a Cryogenic Far-infrared Post-dispersed Polarizing Fourier Transform Spectrometer." *Proc. SPIE 12190*, 2022, <https://doi.org/10.1117/12.2629570>.

Naylor, D. A., Gom, B. G., **Anderson, A. M.**, Huber, A. I., *et. al.* "Development of a Cryogenic Far-infrared Post-dispersed Polarising Fourier Transform Spectrometer." *2022 47th International Conference on IRMMW-THz*, 2022, <https://doi.org/10.1109/irmmw-thz50927.2022.9896015>.

Christiansen, A. J., Naylor, D. A., **Anderson, A. M.**, Gom, B. G. "A Cryogenic FMCW Range-resolved Laser Interferometer: Challenges and Applications." *Proc. SPIE 12008*, 2022, <https://doi.org/10.1117/12.2607894>.

Anderson, A. M., Naylor, D. A., Huber, A. I., Gom, B. G., *et. al.* "Development of a Far-infrared Grating Spectrometer for a Post-dispersed Fourier Transform Spectrometer." *OSA Optical Sensors and Sensing Congress 2021 (AIS, FTS, HISE, SENSORS, ES)*, 2021, <https://doi.org/10.1364/fts.2021.fm5c.4>.

Huber, A. I., Naylor, D. A., **Anderson, A. M.**, Gom, B. G., *et. al.* "Preliminary Results of Line Extraction from a Far-infrared Post-dispersed Polarizing FTS." *OSA Optical Sensors and Sensing Congress 2021 (AIS, FTS, HISE, SENSORS, ES)*, 2021, <https://doi.org/10.1364/fts.2021.fm5c.5>.

Naylor, D. A., Gom, B. G., Huber, A. I., **Anderson, A. M.**, *et. al.* "Development of a Cryogenic Far-infrared Post-dispersed Polarizing Fourier Transform Spectrometer: A Demonstrator for the SPICA SAFARI Instrument." *Proc. SPIE 11453*, 2020, <https://doi.org/10.1117/12.2561100>.

Anderson, A. M., Naylor, D. A., Huber, A. I., Gom, B. G., *et. al.* "Development of a Cryogenic Far-infrared Grating Spectrometer for a Post-dispersed Fourier Transform Spectrometer." *2020 45th International Conference on IRMMW-THz*, 2020, <https://doi.org/10.1109/irmmw-thz46771.2020.9370473>.

Huber, A. I., Naylor, D. A., **Anderson, A. M.**, Gom, B. G., *et. al.* "Development of a Fourier Transform Spectrometer to Assess Performance of an Analogue of the SPICA SAFARI Instrument." *2020 45th International Conference on IRMMW-THz*, 2020, <https://doi.org/10.1109/irmmw-thz46771.2020.9370568>.

CONFERENCE PRESENTATIONS

Extremely Precise Radial Velocities 6

Porto, Portugal

June 30th - July 4th, 2025

- *Poster presentation:* "Testing the HARPS3 Data Reduction Pipeline with Synthetic Data to achieve EPRV."

Exoplanets 5

Leiden, The Netherlands

June 16th - 21st, 2024

- *Poster presentation:* "Investigating sources and propagation of errors in EPRV measurements with HARPS3."

2024 UK Exoplanet Meeting

Birmingham, UK

April 9th - 11th, 2024

- *Poster presentation:* “HARPS3 and the Terra Hunting Experiment.”

2023 UK Exoplanet Meeting

London, UK

August 8th - 10th, 2023

- *Poster presentation:* “Minimising noise contributions to achieve EPRV with HARPS3 and the THE.”

2023 UK National Astronomy Meeting

Cardiff, Wales, UK

July 3rd - 7th, 2023

- **Oral presentation:** “Update on the Terra Hunting Experiment and HARPS3 for the Isaac Newton Telescope.”

SPIE Astronomical Telescopes + Instrumentation 2022: Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI

Montréal, Québec, Canada

July 17th - 22nd, 2022

- *Poster presentation:* “Determining the efficiency of a cryogenic far-infrared diffraction grating spectrometer used as a post-dispersing module for a high-resolution spectrometer.”

OSA Optical Sensors and Sensing Congress Fourier Transform Spectroscopy 2021

Virtual

July 19th - 23th, 2021

- **Oral presentation:** “Development of a far-infrared grating spectrometer for a post-dispersed Fourier transform spectrometer.”

2020 45th International Conference on Infrared, Millimeter, and Terahertz Waves

Virtual

November 9th - 13th, 2020

- **Oral presentation:** “Development of a Cryogenic Far-infrared Grating Spectrometer for a Post-dispersed Fourier Transform Spectrometer.”